Communication Module – Functional Workflow

Project: TaskFlow – Intelligent Task Allocation & Tracking System
Module: Communication (menu: Communication Management) · **Child Modules:** Notifications (incl. deadline reminders engine), Mail
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1. Module Overview

The Communication module is how TaskFlow tells people about their work and how they talk to each other inside the application. It has two parts: **Notifications** are short in-app alerts, for example a new task, a reassignment, a comment, or a deadline reminder. **Mail** is a private internal mailbox for messages between TaskFlow users. A background **reminder engine** checks open tasks every minute. It sends deadline, due-today and overdue reminders as notifications and, when mail delivery is switched on, as Inbox mails from "TaskFlow".

2. Purpose / Business Value

- Nobody misses an assignment, reassignment, comment or approaching deadline, because TaskFlow raises the alert automatically.
- Deadline reminders run on their own, on a schedule set per task (first and second reminder, due today, overdue daily or weekly). Each reminder arrives once only, no matter how often the engine runs.
- Staff can write to each other about their work without leaving TaskFlow or using outside email. Each mailbox is private to its owner.
- Every user sees a single count of what is still unread: the bell in the header for notifications and a badge on the **Mail** menu item.

3. Who Can Access This Module

Role	Can See Module	Can Use Module	Access Level (Own / Team / Organization)
Administrator	✅	✅	Own (own notifications and own mailbox only)
Manager	✅	✅	Own (own notifications and own mailbox only)
Employee	✅	✅	Own (own notifications and own mailbox only)
Custom Role	Configurable	Configurable	Own, as granted

No role, including Administrator, can read another person's notifications or mail. Every list and every action works on the signed-in user's own records. If a record belongs to someone else, the screen reports "not found".

4. Screens in This Module

#	Screen Name	Menu Path	Purpose	Visible To
1	Notifications (Notification Center)	Communication Management → Notifications (also the bell icon in the header)	Read, filter and mark as read your own notifications, and open the related task	Roles holding `notifications.view` (Administrator, Manager, Employee by default)
2	Mail	Communication Management → Mail (shows an unread badge)	Read, write, reply to, forward and organise internal mail, with attachments	Roles holding `mail.read` (Administrator, Manager, Employee by default)
–	Compose panel (inside Mail)	Mail → **Compose**, **Reply**, **Reply all**, **Forward**, or open a draft	Write a new mail, reply or forward one, or continue a saved draft	Roles holding `mail.compose`

> The original BRD Screen 14 "Email Configuration" (SMTP settings) is marked **SUPERSEDED** in the requirements. It was replaced by the internal Mail screen. The one setting that remains, "deliver reminders to Mail", lives on the Reminder Settings screen (Settings module).

5. Child Modules

#	Child Module	Purpose	Visible To	Document
5.1	Notifications (incl. deadline reminders engine)	In-app alerts about tasks and account events, the header bell count, and the automatic deadline, due-today and overdue reminders	Roles holding `notifications.view` (Administrator, Manager, Employee by default)	`05.1-Notifications.md`
5.2	Mail	Private internal mailbox: folders, search, compose, reply, forward, drafts, flags, trash and attachments	Roles holding `mail.read` (Administrator, Manager, Employee by default)	`05.2-Mail.md`

The child documents are in the `05-Communication/` folder next to this file.

6. Permission Summary (Module Level)

Permission Key	Description	Administrator	Manager	Employee	Custom Role
`notifications.view`	View notifications: open the notification centre and the header bell count	Granted	Granted	Granted	Configurable
`notifications.manage`	Mark notifications read: mark one or all as read	Granted	Granted	Granted	Configurable
`mail.read`	Read mail: open the inbox, folders and messages; Mail menu badge	Granted	Granted	Granted	Configurable

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Permission Key	Description	Administrator	Manager	Employee	Custom Role
`mail.compose`	Write and send mail: compose, save drafts, reply and forward, recipient lookup	Granted	Granted	Granted	Configurable
`mail.organize`	Organize mail: star, mark important, move to trash, restore and delete	Granted	Granted	Granted	Configurable
`mail.attachments`	Use mail attachments: upload, download and remove attachments	Granted	Granted	Granted	Configurable

> All six keys are granted by default to Administrator, Manager and Employee. They were also granted to every role that existed when the permissions were first seeded. A new custom role starts with none of them, and an administrator can grant them (or copy them from another role) in Role Configuration. The reminder engine itself has no user permission. Its settings are guarded by `settings.reminders.manage` in the Settings module. The `notifications.*` keys are detailed in `05.1-Notifications.md` and the `mail.*` keys in `05.2-Mail.md`.

7. Related Modules

- **Monitoring (Task Management, Comments, Reassignment)** – creating, reassigning, commenting on and logging hours on tasks creates notifications. Comments and reassignments also create mails. The task's reminder intervals and its notification recipients drive the reminder engine.
- **Settings (Reminder Settings, Application Settings)** – Reminder Settings switches reminders, the deadline-day reminder and "deliver reminders to Mail" on or off. Application Settings controls the in-app notifications toggle and the application timezone used for reminder days.
- **Authentication** – an MFA reset by an administrator and a password reset with the authenticator each send a System notification to the account owner.
- **TaskFlow Assistant (TFA)** – the assistant can find users by name for a mail and draft a mail that the user reviews and confirms before it is sent.
- **Settings → Role Configuration** – grants or removes the six Communication permission keys for custom roles.
- **Between the child modules** – the reminder engine (Notifications) also delivers reminders into the Mail Inbox as mails from "TaskFlow" when "Also deliver reminders to Mail" is on.

8. Expected Outcome

Every user receives timely, non-duplicated alerts about their tasks and deadlines in one Notification Center, and sees at a glance what is unread. They can also exchange private mail with colleagues inside TaskFlow, including replies, forwards, drafts and attachments, without anyone else being able to read their notifications or mailbox.

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